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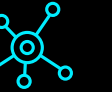
1ST ASSESSMENT

TRACK 2:

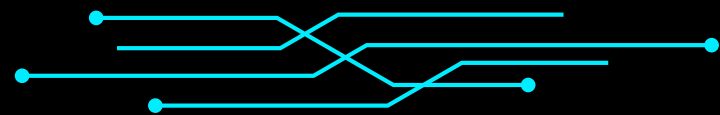
AUTOMATED POTHOLE DETECTION AND
SEVERITY GRADING USING TRANSFER
LEARNING

GROUP MIXUE

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problem STATEMENT



Potholes pose serious risks to road safety and increase vehicle maintenance costs. Current methods, including manual inspection and basic AI models, are often slow and inefficient. Existing deep learning approaches, such as ResNet, can only detect the presence of potholes but cannot provide detailed information like location, size, or severity.





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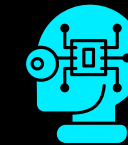
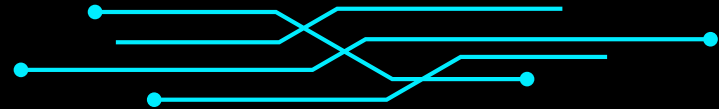
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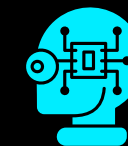
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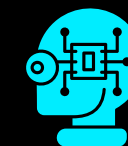
OBJECTIVE



To develop an integrated IoT detection and classification system



To establish a cloud-based mapping and real-time visualisation platform



To evaluate system effectiveness for enhanced road safety

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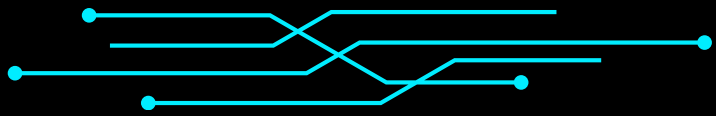
SCOPE

- **YOLOv8**
- **Severity Classification**
- **Weather-Robust False Positive Filtering**
- **Real-Time Cloud-Edge Mapping**



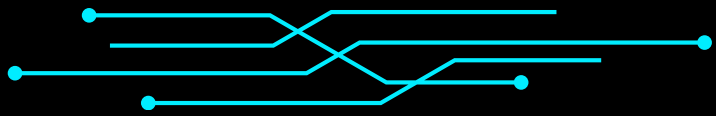
LITERATURE REVIEW

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1) <u>Smart Pothole Detection and Reporting System: Real- Time Localization and Automated Alerts for Efficient Road Maintenance</u>	• Proposed system architecture is design for n layer structure; Sensing layer, processing layer, mapping layer, communication layer • For major potholes, the system triggers an Automated Reporting sequence, instantly raising alerts and notifying all concerned authorities
2) <u>Pothole Detection and Reporting System Using Deep Learning</u>	• To capture additional information on potholes in the specified collection region, a questionnaire survey was implemented where system users could provide more context on the state of the road based on their perspective.
3) <u>Road Pothole Detection and Reporting System</u>	• A seamless user experience was produced by integrating contemporary technologies like React for the frontend and Spring Boot for the backend services
4) <u>PotholeVision: An Automated Pothole Detection and ReportingSystem using Computer Vision</u>	• Proposed severity estimation is based on the number of potholes identified within an image.
5) <u>Pothole Detection using CNN and YOLO v7 Algorithm</u>	• a lesser version of YOLO, which would limit accuracy, and this was the issue that was found to be the most widespread fault across all of the mentioned publications.



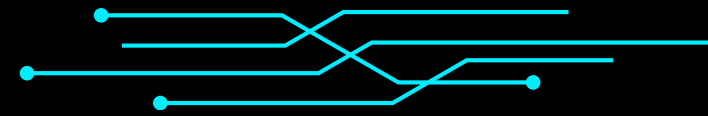
LITERATURE REVIEW

6) Vision-Based Pothole Detection and Reporting System for Smart Cities (Leveraging YOLOv8-Seg and Predictive Maintenance Models)	• Leveraged YOLOv8-Seg to create precise "masks" of potholes, allowing for much more accurate area calculation than standard bounding boxes . • Integrated detection with a predictive maintenance model and a cloud-based reporting system that attaches GPS coordinates to each localized road hazard.
7) Pothole Severity Classification Using Structural Features Extracted from Segmented Images	• Used K-Means clustering and a Decision Tree to classify potholes into severity levels with 98.9% accuracy, providing the mathematical basis for severity grading.



LITERATURE REVIEW

8) Pothole and Wet Surface Detection Using Pretrained Models and ML Techniques	• Addressed real-world environmental gaps by distinguishing between actual potholes and wet road surfaces (puddles) to reduce false positives . • Utilized pretrained models like MobileNet and ResNet50 paired with Random Forest classifiers to maintain high precision in adverse weather conditions.
9) YOLOv8-seg-CP: a lightweight instance segmentation algorithm for chip pad based on improved YOLOv8-seg model	• Introduced a "StarNet" backbone and a shared convolution segmentation head to significantly reduce the number of parameters and computational load. • Developed specific modules (C2f-Star) to enhance the extraction of small targets, which is critical for detecting small-severity potholes at high speeds.

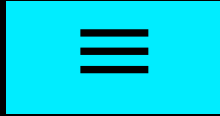
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LITERATURE REVIEW

10) Automatic Detection of Pothole Distress in Asphalt Pavement Using Improved Convolutional Neural Networks

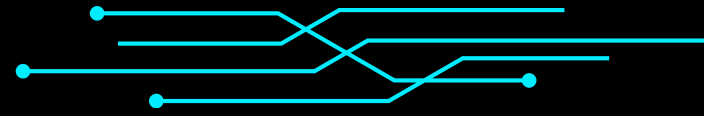
- added "Attention Mechanisms" that act like a zoom lens, helping the AI focus only on the road surface and ignore distracting backgrounds like trees or buildings.
- uses "Multi-Scale Feature Fusion," which allows it to accurately detect both large potholes nearby and very small potholes further down the road.



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